

Low triglyceride levels are associated with increased risk of immune-related adverse events in patients receiving immune checkpoint inhibitors

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ABSTRACT

Immune checkpoint inhibitors (ICI) have revolutionized cancer therapy by enhancing anti-tumor immune responses, yet their use can lead to immune-related adverse events (irAE), including neurological complications. Despite their clinical relevance, predictive biomarkers for irAE remain scarce, and early identification of at-risk patients is a major unmet need. In this prospective study, 200 patients undergoing ICI therapy were enrolled, of whom 59 underwent longitudinal metabolomic profiling at baseline, three months, and six months. Thirty-two patients who developed irAE were compared to 27 age- and sex-matched individuals without irAE. Multivariate analyses, including Principal Component Analysis (PCA) and Partial Least Squares Discriminant Analysis (PLS-DA), revealed distinct metabolomic signatures differentiating the two groups. Notably, baseline levels of triglyceride 20:0_34:1 were significantly lower in irAE(+) patients. In female patients, additional triglyceride species—20:1_34:2, 20:2_34:2, and 20:2_34:3—were also reduced prior to therapy and showed increases within three months of ICI initiation. These findings suggest that specific triglyceride species may serve as early biomarkers for irAE risk, particularly in female patients. The observed dynamic changes point to a potential link between lipid metabolism and immune-related toxicity, supporting the integration of metabolomic profiling into future strategies for risk stratification and personalized monitoring in cancer immunotherapy.

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1. Introduction

Immune-checkpoint inhibitors (ICI) have revolutionized oncological treatment within the last decade and are currently considered as standard of care in a plethora of malignancies.¹ ICI are monoclonal antibodies that inhibit particular immune-checkpoint receptors like cytotoxic T-lymphocyte-associated protein 4 (CTLA-4), programmed cell death protein 1 (PD-1), or its ligand (PD-L1). Therefore, ICI suppress inhibitory pathways and increase mainly CD8⁺ T-cell activity, augmenting the antitumor response, ultimately terminate cancer cells immune evasion.^{2,3} However, this nonspecific immune activation could also be directed against endogenous tissue and consequently trigger so called immune-related adverse events (irAE). IrAE are potentially serious autoimmune adverse events of ICI-therapy that can affect almost every organ. Herein, hypothyroidism, colitis, dermatitis, hepatitis, pneumonitis and hypophysitis are well-

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known types of irAE.⁴ Their severity can range from marginal to life threatening.^{5,6} Recently, neurological adverse events like encephalitis or peripheral neuropathy are of special interest, as they are relatively rare but often associated with increased morbidity and also mortality.⁷ Regardless of which organ is affected, rapid and effective initiation of immunosuppressive therapy is key. Predominantly, irAE associated inflammatory reactions are treated with steroids, as well as with interruption or cessation of ICI therapy.³

Therefore, intensive monitoring of patients undergoing ICI therapy is mandatory for early irAE treatment and detection, particularly because early immunosuppressive therapy seems to be associated with better outcomes of irAE.^{8–10} However, clinical monitoring alone is often insufficient to timely detect serious side effects or prevent their occurrence.¹¹ Despite extensive research, no biomarker for irAE prediction has yet been prospectively validated. Recently, however, our group identified MCP-1 as a promising serum-biomarker for neurological irAE.⁷

This current retrospective analysis, utilizing data from a prospective register of ICI-treated patients, sought to uncover potential predictive biomarkers for irAE. For the first time, we focused on metabolomic serum patterns in patients with and without irAE, opening new avenues in biomarker exploration.

2. Patients and methods

2.1. Study design and patient selection

Since December 2019, an interdisciplinary Immune-Oncology-Working Group (ICOG) at Hannover Medical School (MHH) established a prospective cohort of oncological patients undergoing ICI treatment. Herein, the Skin-Cancer-Center, the department of Hematology, Hemostasis, Oncology and Stem Cell Transplantation, the department of Pneumology, and Gastroenterology recruit patients with ICI treatment.⁷

Required inclusion criteria were an oncological diagnosis, a treatment with immune-checkpoint inhibitors, age over 18 years and written informed consent. As per study protocol, patients are observed for 6 months with seven study visits at defined points in time: (I) before ICI-therapy (baseline); (II) three to four weeks after baseline (FU1); (III) three to four weeks after FU1 (FU2); (IV) three to four weeks after FU2 (FU3); (V) three to four weeks after FU3 or three months after baseline (FU4); (VI) three to four weeks after FU4 (FU5); (VII) six months after baseline (end-of-study-visit (EOS)) (Supplemental Figure S1).

The focus of the study was on recognizing irAE as early as possible to initiate further diagnostics and treatment to prevent complications and long-term sequelae of irAE. The documented adverse events were categorized according to their severity ranging from grade 1 (mild) to grade 5 (lethal) based on the Common Terminology Criteria for Adverse Events (CTCAE) version 5.0. In this study, only those symptoms with a very high probability of being associated with the immunotherapy and documented in the patients' clinical records were considered. Subsequently, 59 patients were selected for metabolomic serum analysis based on age- and sex-matching. Patients with irAE were manually matched to individuals without irAE using a pairwise approach to ensure demographic comparability and representative coverage across tumor types and therapies. The study protocol was approved by the local Ethics Committee at Hannover Medical School (No. 8685_BO_K2019) following the Declaration of Helsinki.

2.2. Targeted metabolomics analysis

For the targeted metabolomics analysis, 59 patients were chosen from the total cohort ($n = 200$) and subgrouped. IrAE(+) patients consisted of 32/59 patients who developed documented irAE (CTCAE 1: $n = 5$, CTCAE 2: $n = 11$, CTCAE 3: $n = 16$; Supplemental Table S1). The control group (irAE(-)) included 27 age- and sex-matched patients who did not suffer from autoimmune side effects. From each of these 59 patients, serum samples that were taken at three defined points in time (before therapy start (baseline), after 3 months (FU3 or FU4) and after 6 months (end of study)) were analyzed. All serum samples were stored at -80°C and analyzed in duplicate using a targeted metabolomics kit (MxP Quant 500 kit, Biocrates Life Science AG, Innsbruck, Austria). This kit enables quantitative measurements of 630 analytes, comprising various small molecules (alkaloids ($n = 1$), amine oxides ($n = 1$), amino acids ($n = 20$), amino acid related ($n = 30$), bile acids ($n = 14$), biogenic amines ($n = 9$), carbohydrates and related ($n = 1$), carboxylic acids ($n = 7$),

cresols ($n = 1$), fatty acids ($n = 12$), hormones ($n = 4$), indoles and derivatives ($n = 4$), nucleobases and related ($n = 2$), vitamins and cofactors ($n = 1$), and lipids (acylcarnitines ($n = 40$), phosphatidylcholines ($n = 76$), lysophosphatidylcholines ($n = 14$), sphingomyelins ($n = 15$), ceramides ($n = 28$), dihydroceramides ($n = 8$), hexosylceramides ($n = 19$), dihexosylceramides ($n = 9$), trihexosylceramides ($n = 6$), cholesteryl esters ($n = 22$), diglycerides ($n = 44$), and triglycerides ($n = 242$)). Metabolite extraction and analyses were conducted following the manufacturer's protocol (<https://biocrates.com/mxp-quant-500-kit>, accessed on 05 June 2023). Metabolite concentrations were measured on an AB SCIEX 5500 QTrapTM mass spectrometer (AB SCIEX, Darmstadt, Germany).

Data processing: After normalization and pre-processing of the data, MetIDQTM software (Biocrates) was used for peak integration and calculation of metabolite concentrations. All analytes that were above the limit of detection (LOD) in $\geq 80\%$ of patients of at least one group were selected for further investigation as applied previously.^{12,13} Values $< \text{LOD}$ were replaced by the reference value provided by the manufacturer ($\text{LOD}/2$). Missing values were replaced by sample-wise k-nearest neighbors' algorithm using MetaboAnalyst 5.0.

2.3. Statistical analysis

The metabolite concentration data were entered into the Metaboanalyst 5.0 or 6.0 software for uni- and multivariate analyses, including t-test, where the Benjamini-Hochberg correction was applied with a false discovery rate (FDR) of 0.1 to account for multiple testing. A fold change (FC) threshold of > 1.5 was set for differential abundance analysis, which was visualized as volcano plot. Univariate receiver operating characteristic (ROC) curve analysis was employed to evaluate the identified metabolites' performance as possible biomarkers. Differential abundance analysis and ROC curve analysis were performed using open source software MetaboAnalyst 5.0 (<http://www.metaboanalyst.ca>).

The abbreviations used for phosphatidylcholines (PC), triglycerides (TG), sphingomyelins (SM), hexosylceramides (HexCer) and ceramides (Cer) correspond to the Biocrates nomenclature (<https://biocrates.com/wp-content/uploads/2022/02/biocrates-Quant500-list-of-metabo-lites-v6-2022.pdf>).

To investigate whether metabolite concentrations were associated with CTCAE grade, we performed a linear trend analysis across CTCAE grades within the female baseline subgroup ($n = 7$). Metabolite values were $\log_{10}(x + 1)$ – transformed, and separate linear regression models were fitted for each selected metabolite, using CTCAE grade as a numeric predictor. The analysis was performed in R (version 4.4.0), and p-values were adjusted for multiple comparisons using the Benjamini-Hochberg FDR correction.

Further statistical analyses were performed using GraphPad prism (version 10) and SPSS (version 27). For group comparison, unpaired t-test was used for continuous data. For comparison of nominal data, the Mann-Whitney U-test was applied. Two-way ANOVA was used to compare the differences in metabolite concentrations in the two groups at three different time points. A multivariate logistic regression analysis was conducted using SPSS Version 27 (IBM Corp., Armonk, NY) to evaluate the predictive value of the previously identified serum metabolites for the development of irAE. The primary outcome variable was the occurrence of irAE, with several predictor variables considered for their potential to independently predict irAE. Results with $p < 0.05$ were considered statistically significant.

3. Results

3.1. Patients' characteristics of the total cohort

At the time of evaluation of the cohort, 200 patients had completed their end-of-study visit (6 months after therapy initiation) and were eligible for data analysis. Details of the patient characteristics are shown in Table 1 and Supplemental Table S1. In this cohort, nearly half of all patients (92/200; 46%) developed ICI-associated irAE and 18% (36/200) even suffered from more than one irAE. Taken together, 140 adverse events occurred in 92 patients. Significantly more patients in the group with irAE had malignant melanoma ($p = 0.0289$), while significantly fewer suffered from lung carcinoma ($p = 0.0383$). Regarding the used ICI, significantly more patients with irAE were treated with a combination of nivolumab and ipilimumab ($p = 0.0013$). The most common adverse events affected the thyroid gland with 17% of all irAE appearing

as thyroiditis or hypothyroidism. Other typical sites of irAE-manifestation in the cohort were the gastro-intestinal tract (14%), liver (11%), and skin (11%). According to CTCAE, 21% of the patients with adverse events suffered from mild symptoms of grade 1 and 36% showed moderate side effects which were classified as grade 2. Higher-grade irAE (CTCAE 3) which required hospitalization occurred in 39 patients (42%). One patient (1%) developed a life-threatening immune-related complication (CTCAE grade 4). Specific neuromuscular adverse events (nAE) were detected in 9% (18/200) of all patients. The most common neurological side effects were polyneuropathy and myopathy. In two severe cases, the patients developed autoimmune encephalitis and myositis-myasthenia-overlap-syndrome, respectively.

3.2. Patients' characteristics of the sub-cohorts selected for metabolomics analysis

The sub-cohort in which targeted metabolomics analysis was performed consisted of 59 selected patients from the total cohort of 200 evaluable patients. Among them, 32 patients had suffered considerable irAE (irAE (+)) and were compared to an intentionally age- and sex-matched group of 27 patients (control group) without irAE (irAE (-)). Regarding median age and sex of the patients, there were no significant differences between irAE (+) and irAE (-) patients (Table 1).

Most of the irAE (+) patients received nivolumab (50%) or the combination of nivolumab and ipilimumab (31%). The distribution of irAE resembled the total cohort with the thyroid being the most frequently affected organ (24%) (Supplemental Table S1). According to CTCAE, the vast majority of the irAE (+) individuals suffered from moderate (grade 2) ($n = 11$) or higher-grade (grade 3) ($n = 16$) adverse events. The two groups, irAE (+) and irAE (-), did not differ significantly with respect to baseline parameters (Table 1). One noticeable difference was – as in the total cohort – that significantly more patients in the irAE (+) group were treated with a combination therapy of nivolumab and ipilimumab compared to the irAE (-) group ($p = 0.0077$).

3.3. Metabolome analysis

3.3.1. Comparison of metabolome profiles of patients with and without irAE at baseline (whole cohort)

Of the 519 metabolites (Supplemental Figure S2) in the patients' serum samples that were suitable for analysis, three metabolites (cholic acid (CA), triglyceride (TG (20:0_34:1)), and diglyceride (DG (16:1_18:1))) showed a considerable difference in baseline-concentrations between irAE(+) and irAE(-)

Table 1. Patient characteristics of the total cohort and the selected subgroups with irAE and without irAE.

Parameters	Total cohort ($n = 200$)			Selected metabolomics cohort ($n = 59$)		
	irAE(+) ($n = 92$)	irAE(-) ($n = 108$)	p-value	irAE(+) ($n = 32$)	irAE(-) ($n = 27$)	p-value
Median age at start of ICI in years (IQR)	61 (53–72)	65 (56.5–73)	0.1158	61.5 (53–72)	62 (52–70)	0.8665
Sex	62 (67.4%)	62 (57.4%)	0.1883	24 (75%)	15 (55.6%)	0.1682
male	30 (32.6%)	46 (42.6%)		8 (25%)	12 (44.4%)	
female						
Tumor entities	64 (69.6%)	58 (53.7%)	0.0289	24 (75%)	20 (74.1%)	0.9999
malignant melanoma	10 (10.9%)	24 (22.2%)	0.0383	3 (9.4%)	1 (3.7%)	0.6175
lung cancer	12 (13%)	14 (13%)	> 0.9999	3 (9%)	4 (14.8%)	0.6917
renal cell carcinoma	5 (5.4%)	7 (6.5%)	0.7766	2 (6.25%)	2 (7.4%)	0.9999
head and neck cancer						
HCC	1 (1%)	3 (2.8%)	0.6261	0 (0%)	0 (0%)	0.9999
others [#]	0 (0%)	2 (1.9%)	0.5007	0 (0%)	0 (0%)	0.9999
Immunotherapy*	43 (46.7%)	51 (47.2%)	0.8864	16 (50%)	18 (66.7%)	0.2905
nivolumab	16 (17.4%)	29 (26.9%)	0.0927	5 (15.6%)	5 (18.5%)	0.9999
pembrolizumab	28 (30.4%)	12 (11.1%)	0.0013	10 (31.3%)	1 (3.7%)	0.0077
nivolumab+ipilimumab	3 (3.3%)	7 (6.5%)	0.3421	1 (3.1%)	0 (0%)	0.9999
atezolizumab	1 (1.1%)	2 (1.9%)	> 0.9999	0 (0%)	1 (3.7%)	0.4576
avelumab	0 (0%)	2 (1.9%)	0.4996	0 (0%)	0 (0%)	0.9999
durvalumab	0 (0%)	2 (1.9%)	0.4996	0 (0%)	2 (7.4%)	0.2051
cemiplimab	1 (1.1%)	0 (0%)	0.4670	0 (0%)	0 (0%)	0.9999
ipilimumab						

The subgroups were compared with the t-test. P-values < 0.05 were considered statistically significant and are shown in bold. *In three patients within the non-irAE group, the type of immunotherapy is not known. Abbreviations: ICI: immune-checkpoint inhibitors, irAE: immune-related adverse events, IQR: interquartile range, HCC: hepatocellular carcinoma. [#]Other tumor entities consisted of neuroendocrine carcinoma and transitional cell carcinoma of the bladder.

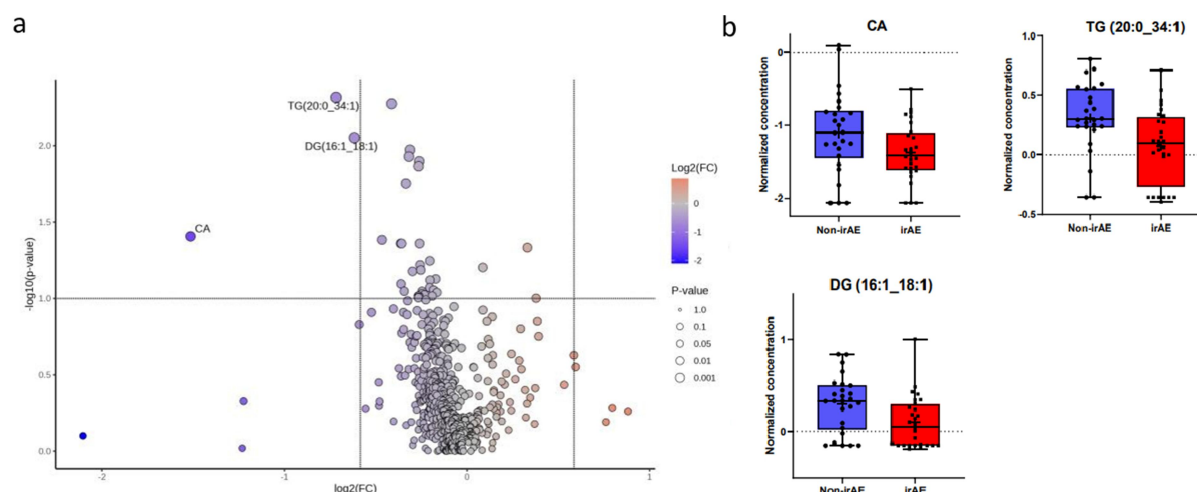


Figure 1. Serum metabolome profile of whole cohort at baseline (comparison of irAE(+) and irAE(-)). a: Volcano Plot: Each point represents a metabolite, blue represents the potentially “down-regulated” metabolites compared with Non-irAE group (raw p-value < 0.1, and fold change (FC) > 1.5). b: Box-plots of normalized concentration differences between the irAE(+) and irAE (-) group for the 3 identified metabolites: cholic acid (CA), TG (20:0_34:1), and DG (16:1_18:1). The black dots indicate individual sample concentrations. Mean (black line) and median (+) are marked, the boxes represent 95% confidence-interval of each group, defined as $\pm 1.58 \cdot \text{IQR} / \sqrt{n}$. Vertical black lines represent range of individual sample concentrations. Due to preceding data normalization applied, we received a negative scale on the Y-axis for some data points. Abbreviations: irAE: immune-related adverse events.

groups according to differential abundance analysis, if the entire cohort was considered and no distinction between sexes was made (Figure 1a). The concentration of the identified metabolites was significantly reduced in the serum of irAE (+) patients at baseline (Figure 1b). In the multivariate logistic regression analysis, higher concentrations of DG (16:1_18:1) and TG (20:0_34:1) were identified as independent predictive factors, each significantly associated with a reduced risk of developing irAE (Supplemental Table S2).

3.3.2. Comparison of metabolomic profiles of female patients with and without irAE at baseline

Given the well-documented differences between men and women in the development of autoimmune diseases – and the likelihood that these differences extend to irAE – a sex-specific analysis was incorporated into the study. Within the female patients, irAE (+) and irAE (-) subgroups did not differ in terms of age or underlying tumor disease (Supplemental Table S3). Twenty out of 519 potential serum metabolites (Supplemental Figure S2) were significantly downregulated in irAE (+) female patients (Figure 2b). The heat map of the top 25 metabolites revealed that nearly all were down-regulated in irAE (+) and up-regulated in irAE (-) (Figure 2a). Except for one diglyceride, all metabolites with significant concentration differences were triglycerides. ROC curve analysis, used to identify potential biomarkers, showed an area under the curve (AUC) of > 0.8 for all 20 metabolites identified by FDR-corrected t-test (Figure 2c). Triglyceride 20:1_34:2 at baseline showed the most significant concentration difference between the groups (Figure 2d). Combining results from differential abundance and univariate ROC analysis, Triglyceride 20:1_34:2 emerged as the most promising biomarker, with an AUC of 1.0 (Figure 2e).

In line with our findings within the female group, principal component analysis showed a convincing separation of the two groups at baseline (Figure 3a). In addition to the triglycerides identified by univariate analysis, three metabolites with increased concentrations in the irAE (+) group, namely glycodeoxycholic acid (GDCA), p-cresol sulfate (p-Cresol-SO₄), and deoxycholic acid (DCA), were also important in driving this separation (Figure 3b). Exploratory analysis within the female subgroup revealed no statistically significant association between CTCAE grade (CTCAE 2: $n = 3$; grade 3: $n = 4$) and log-transformed concentrations of the top 20 triglycerides after FDR correction (all $p > 0.5$), despite nominal significance for individual metabolites (e.g., TG (20:3_34:3), $p = 0.028$) (Supplemental Table S5).

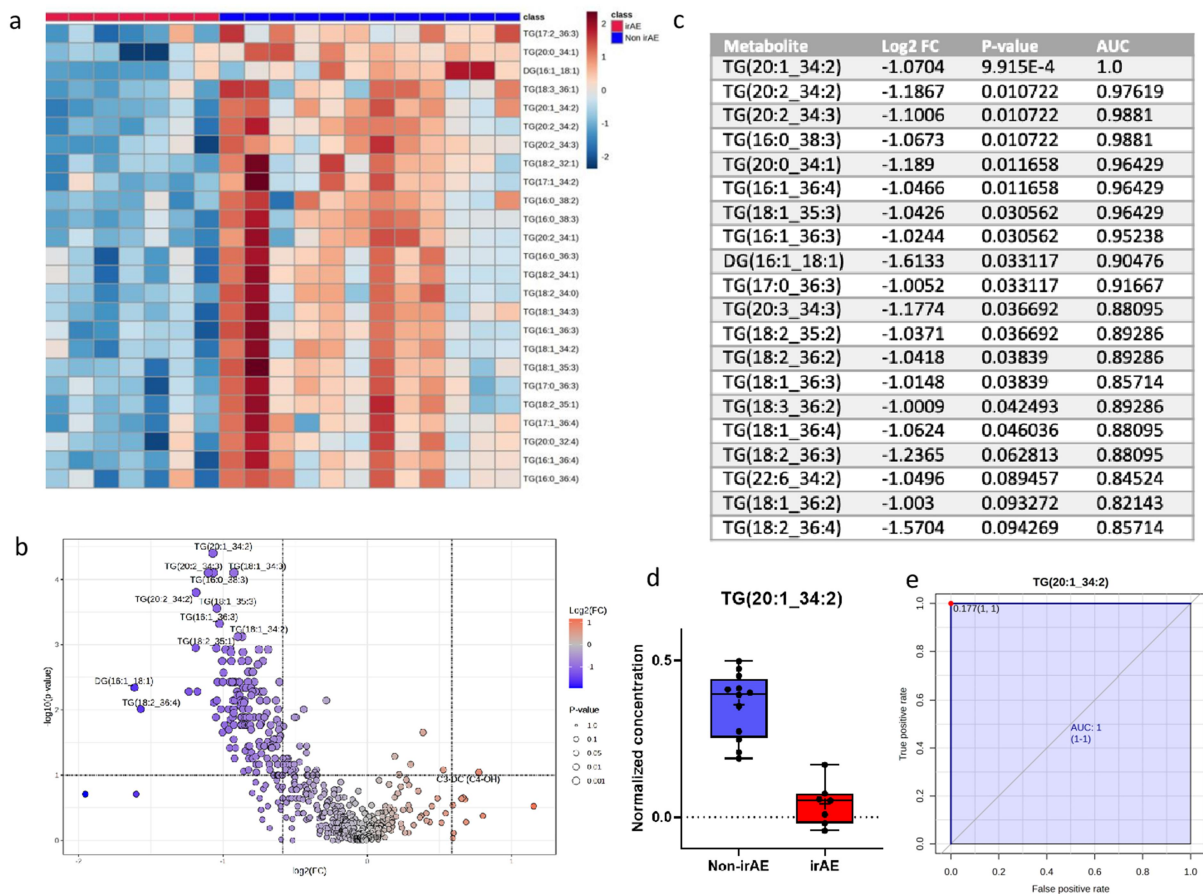


Figure 2. Serum metabolome profile of female patients prior to start of immune-checkpoint inhibitor therapy. a: Heatmap of Top 25 metabolites (t-test, using Euclidean clustering, Ward method). Twenty of them were significantly decreased in the irAE subgroup in the FDR adjusted T test (see c). Each colored cell on the map corresponds to a concentration, with samples in rows and compounds in columns. Red and blue colors indicate positive and negative correlations. Baseline blood was not available from one female patient with irAE, which is why only 7/8 female patients with irAE could be included in the analysis. b: Volcano Plot: Each point represents a metabolite, blue dots represent potentially down-regulated metabolites (145/519 significantly down-regulated) in females with irAE compared with the irAE (-) group and red dots represent potentially up-regulated metabolites (1/519 significantly up-regulated); (FDR-adjusted p-value < 0.1, and fold change (FC) > 1.5). c: The table shows the statistical parameters from the differential abundance analysis and the ROC analysis for selected metabolites. The 'Top 20' metabolites are listed, sorted according to their FDR-adjusted p-value. d: Exemplary Box-plot visualizing the normalized concentration of the metabolite identified as the best potential biomarker to discriminate irAE from non-irAE in females, triglyceride 20:1_34:2. The black dots indicate individual sample concentrations. Mean (black line) and median (+) are marked, the boxes represent 95% confidence-interval of each group, defined as $\pm 1.58 \cdot \text{IQR} / \sqrt{n}$. Vertical black lines represent range of individual sample concentrations. Due to preceding data normalization applied, we received a negative scale on the Y-axis for some data points. e: Example ROC curve of triglyceride 20:1_34:2 with an AUC of 1.0. Sensitivity is on the y-axis, and specificity is on the x-axis. Abbreviations: AUC: area under the receiver operating characteristic curve; irAE: immune-related adverse events.

To further evaluate the discriminatory potential of metabolite patterns in female patients with irAE, we conducted a multivariate ROC analysis using the PLS-DA – based biomarker module in MetaboAnalyst 6.0. Models with increasing numbers of top-ranked variables ($n = 5$ to $n = 100$) yielded progressively higher AUC values, reaching 0.90 for the top 100 metabolites. However, wide confidence intervals (e.g., CI = 0.571–1.000 for the top 100) indicated limited model robustness, likely due to the small sample size (Supplemental Figure S5).

In male patients, significant baseline concentration differences were observed for triglyceride 22:0_32:4 (TG (22:0_32:4)), and bile acid taurochenodeoxycholic acid (TCDCA) (Supplemental Figure S3A). The concentration of TCDCA was significantly decreased in irAE (+) patients at baseline, while the concentration of TG(22:0_32:4) was increased (Supplemental Figure S3B). The

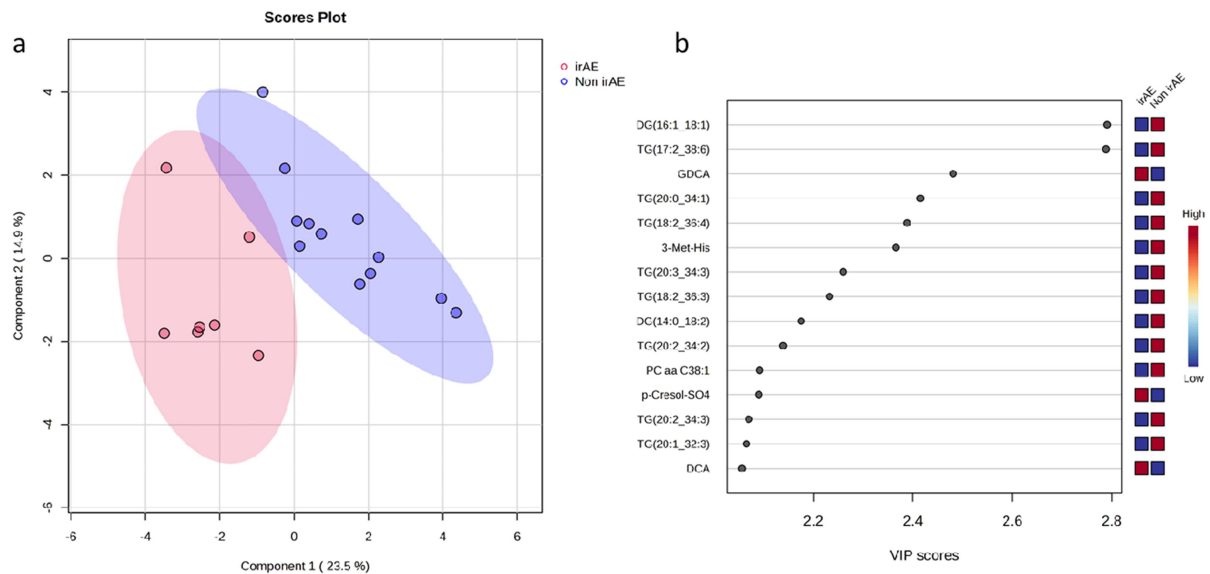


Figure 3. Metabolomic profile in female patients at baseline. a: Partial least squares discriminant analysis (PLS-DA) scores plot showing clear separation between the female patients who developed irAE ($n = 7$) and those who did not ($n = 12$) based on their metabolic profile of selected metabolites. Cross-validation was performed by LOOCV-method: Accuracy 0.78, R^2 0.63, Q^2 0.32. B: Variable influence on projection (VIP) plot showing the metabolites that are most important in driving the separation of the two groups. Abbreviations: irAE: immune-related adverse events.

irAE (+) and irAE (-) subgroups did not differ in terms of age or underlying tumor disease (Supplemental Table S4).

3.3.3. Changes in serum metabolite concentrations over time in female patients

In female patients with irAE, autoimmune side effects occurred an average of 57 days after starting ICI therapy (range: 7–221 days), with most women (75%) experiencing irAE before the three-month follow-up.

At baseline, triglycerides 20:2_34:2 and 20:2_34:3 exhibited the most significant concentration differences between irAE (+) and irAE (-) in the female cohort. In irAE (+), both metabolites exhibited a significant increase in concentration from baseline to the 3-month follow-up, while concentrations in irAE (-) remained unchanged (Figure 4b,c). The majority of the 20 significantly downregulated metabolites in the female irAE (+) patients showed a similar pattern over time (Supplemental Figure S4).

Conversely, triglyceride 20:1_34:2 did not show a significant increase in concentration between baseline and month 3 in irAE (+), serving as an example (Figure 4a).

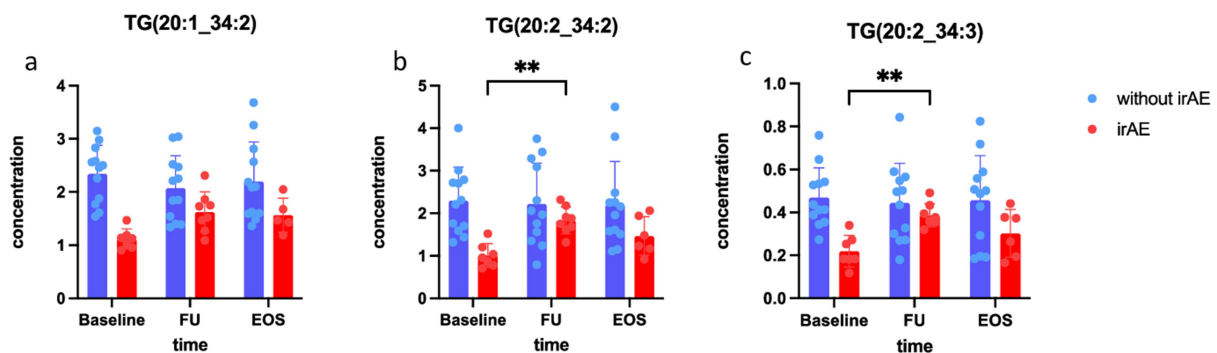


Figure 4. Changes in serum concentrations of the three metabolites with the highest concentration differences at baseline over time. Illustration of absolute concentrations of triglyceride 20:1_34:2 (a), triglyceride 20:2_34:2 (b), and triglyceride 20:2_34:3 (c) at the time of baseline assessment, three-month follow-up, and end-of-study visit at 6 months. ** $p < 0.01$. Abbreviations: EOS: end-of-study visit; FU: follow-up; irAE: immune-related adverse events.

4. Discussion

In the present study, we identified triglycerides as potential biomarkers for predicting irAE in women. Up to now, various potential biomarkers for irAE under treatment with ICI have been discussed in the literature. They range from rheumatic auto-antibodies and changes in blood cell counts^{14–16} to different cytokines, features of the gut microbiome or specific genetic analyses.^{17,18} Most of the methods used, such as Human Leucocyte Antigen (HLA-) genotyping for genetic variability, microRNA and gene profiling, or microbiome analysis, are however not yet established in everyday clinical practice.^{17,19} The findings so far indicate that immunogenomic correlates for the occurrence of irAE during anti-PD-1 immunotherapy do exist. It could for example be demonstrated that an abundance of dendritic cells, a high tumor mutational burden (TMB), and higher counts of CD8+T-cells, mast cells, CD4+T-cells, and CD4+ naive T-cells were accompanied by a higher risk for irAE, with dendritic cells abundance showing the strongest correlation.²⁰ In patients with non-small-cell lung cancer, dyslipidaemia, increased hemoglobin levels, and eosinophil cell count as well as decreased lactate dehydrogenase (LDH) and C-reactive protein (CRP) concentrations prior to ICI-start were considered as predictors for irAE.²¹

Looking for new biomarkers that could provide new insights into the pathogenesis of irAE and ultimately allow better risk stratification, a targeted metabolomics analysis was used in the present study, which also took gender-specific differences into account. For a reliable comparison of the metabolomic profiles in both subgroups (irAE (+) and irAE (-)), a close congruence in the basic patients' characteristics is preferable. Both groups did not differ significantly concerning age, sex, and tumor entities. There were only significant differences in the immunotherapies administered, with irAE (+) patients being treated more frequently with a combination therapy of ipilimumab and nivolumab. This is not unexpected, as a higher risk of side effects is associated with combination therapies.^{5,6,11,18,22,23}

Based on the results of the present study, a connection between lipid metabolism and irAE could be postulated. The targeted metabolome analysis showed decreased baseline concentrations of triglycerides in irAE (+) compared to irAE (-) patients. These results could lead to the hypothesis that certain triglycerides have a protective effect with regard to the development of autoimmune side effects, especially in females. Within the literature, one single study reported the potential connection between lipid metabolism and the occurrence of irAE as yet. Lipidomics analyses performed in patients with non-small cell lung cancer (NSCLC) under ICI treatment identified eight lipids with high sensitivity and specificity as potential biomarkers for the occurrence of irAE.²⁴ The results of this study support our hypothesis about preexisting differences in lipid metabolism and its influence on the occurrence and prediction of irAE. In addition, a link between lipid metabolism (especially triglycerides) and autoimmune diseases in general has already been established. Cai *et al.*, for example, pointed out that lipid metabolism regulates activation, differentiation, and function of CD4+ T cells and is therefore associated with the development of immunological disorders and diseases.²⁵ An earlier publication from 2003 provides a deeper insight into the role of triglycerides in this context. The authors demonstrated that levels of triglycerides were lower in patients with chronic autoimmune diseases such as autoimmune thyroiditis, SLE, rheumatoid arthritis, or atopic dermatitis/asthma.²⁶ A recent study on autoimmune thyroiditis revealed similar results.²⁷ In view of the study evidence, it can be hypothesized that reduced levels of triglycerides in serum may favor autoimmune processes such as ICI-associated autoimmune side effects. Conversely, it can also be postulated that higher concentrations before the start of therapy could be protective with regard to autoimmune side effects. As the changes in the present study were already evident before the start of therapy, the serum triglyceride concentration may also allow pre-therapeutic risk stratification. The variations in triglyceride levels are unlikely to result from differences in comorbidities or nutritional status, as there are no significant differences between the subgroups.

This study was able to contribute new findings with regard to the prediction of irAE by applying a state-of-the-art method. Yet, there are limitations due to the small size of the subgroups, and results should be verified in larger studies. Moreover, there is an uneven gender distribution including fewer women than men. This is particularly relevant as the results of the metabolome analysis showed sex-specific differences.

Within our cohort, female patients exhibited statistically more significant differences in their metabolic profiles with regard to irAE occurrence than men, which might relate to the fact that women have a higher risk of autoimmune diseases in general.^{28,29} With regard to the time course, this study was able to show that

triglyceride concentrations in female irAE (+) patients increased within three to four months after initiation of ICI-therapy. However, their levels did not reach those of the irAE (-) group. After the peak at about three months, the triglyceride concentrations in irAE (+) females remained stable without reaching baseline level. These results suggest the hypothesis that a temporal link between the rise of triglycerides and the occurrence of irAE might exist. Within the female irAE (+) group, autoimmune side effects occurred on average 57 days after the start of ICI therapy, with 75% of women experiencing irAE before the three-month follow-up. However, the subgroup analysis of female patients is limited by a relatively small sample size. Thus, our findings should be interpreted with caution and validated in larger, independent cohorts. As the blood samples in this study were taken at predefined times and not at the exact time of irAE occurrence, it is possible that the samples may have been taken slightly too late and a more significant increase of triglyceride levels would have been recorded at two months. This should be verified in future studies with serum analyses at the time of irAE occurrence.³⁰

The observed association between lower baseline triglyceride levels and the development of irAE suggests a potential link between lipid metabolism and immune regulation. Triglycerides have been shown to influence T-cell function, including activation and cytokine production. Emerging data indicate that lipid metabolic pathways may modulate immune cell behavior in the tumor microenvironment. While our study focused on circulating metabolites, integrating these findings with transcriptomic and immune profiling – such as immune infiltration scores from TCGA datasets or pathway-level analyses via tools like GseaVis or IORB – could further clarify mechanistic underpinnings.^{31,32} Future studies leveraging such multi-omic approaches are warranted.

5. Conclusion

This study identified distinct metabolomic profiles as potential biomarkers for predicting irAE in patients undergoing ICI therapy. Significant differences in specific metabolites, especially triglycerides 20:0_34:1, 20:1_34:2, 20:2_34:2, and 20:2_34:3, were observed between irAE (+) and irAE (-) patients. These metabolites showed dynamic changes over time, suggesting their role in early detection of irAE.

Integrating metabolomic profiling into clinical practice could improve monitoring and management of ICI therapy, allowing for timely interventions and potentially mitigating the severity of irAE. Further validation studies are needed to confirm these findings and standardize their clinical application, paving the way for more personalized and safer cancer immunotherapy.

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N.M. and E.N. contributed equally to the planning, execution, result analysis, and manuscript preparation. L.D., J.T., and L.G.-L. supported data collection and analysis. T.F. and P.I. coordinated the clinical cohort. T.S., P.I. and N. M. conceptualized the study design. S.N. was primarily responsible for the conception and evaluation of the metabolomics analyses. S.S. conducted the metabolomics analyses, while F.P. and W.Z. also contributed to the evaluation of the metabolomic results. B.-A.B., G.B., F.H., T.W., I.v.W., and R.G. contributed patients and were involved in the study design. All authors read and approved the manuscript.

Author contributions

CRediT: **Nora Möhn:** Conceptualization, Data curation, Formal analysis, Investigation, Project administration, Visualization, Writing – original draft; **Emily Narten:** Data curation, Formal analysis, Visualization, Writing – original draft; **Laura Duzzi:** Data curation, Investigation, Writing – review & editing; **Janin Thomas:** Data curation, Investigation, Writing – review & editing; **Lea Grote-Levi:** Data curation, Investigation, Writing – review & editing; **Gernot Beutel:** Resources, Writing – review & editing; **Tabea Fröhlich:** Data curation, Resources, Writing – review & editing; **Benjamin-Alexander Bollmann:** Resources, Writing – review & editing; **Thomas Wirth:** Resources, Writing – review & editing; **Imke von Wasielewski:** Project administration, Resources, Writing – review & editing; **Ralf**

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Disclosure statement

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Data availability statement

The datasets used and analyzed in this study are available from the corresponding author upon reasonable request.

Ethics approval and consent to participate

The study was approved by the local Ethics Committee of Hannover Medical School (MHH) (Ethics application number: 8685_BO_K2019) and adheres to the principles of the Declaration of Helsinki. All participants provided written informed consent.

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